

From Blasius to Falkner-Skan

Solving Nonlinear Boundary-Layer Flows with Verification Certificates

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SIRPY · Problem 11 — One Equation, a Family of Flows

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

From one flow to a family

The previous entry solved the Blasius equation — the laminar boundary layer over a flat plate, with a boundary condition at infinity. This one generalizes it.

$$f''' = -ff'' - \beta(1 - (f')^2), \quad f(0) = 0, \quad f'(0) = 0, \quad f'(\infty) = 1$$

This is the **Falkner–Skan equation**, published by V. M. Falkner and Sylvia Skan in 1930. The single parameter β carries the pressure gradient of the external flow, and changing it changes the physics:

$\beta > 0$ — a favorable pressure gradient. The flow accelerates, and the boundary layer is pressed thin against the wall.

$\beta = 0$ — no pressure gradient. The equation reduces to Blasius: the flat plate.

$\beta < 0$ — an adverse pressure gradient. The flow is decelerating, the boundary layer thickens, and as β decreases the wall shear falls toward zero. At $\beta \approx -0.19884$ it reaches zero and the flow **separates** — the point where a boundary layer detaches from the surface, the mechanism behind stall on a wing.

So β is not a knob. It is the difference between attached flow and separation, and the equation must be re-solved for every value of it.

What we handed the solver

```

for beta in (-0.1, 0.0, 0.5):
    r = solve_bvp(order=3,
        f_str=f"-y*y2 - {beta}*(1 - y1**2)",
        x0=0, x1=float("inf"),
        left_bc={0: 0, 1: 0, 2: "gamma"},
        right_bc={1: 1})

```

A loop. The domain is written as infinite — no truncation point chosen by us — the unknown wall shear is named "gamma", and nothing else is supplied. No initial guess, no mesh, no per-tuning. The same four lines run for every member of the family.

The truncation, certified each time

For each β , SIRPY chose a truncation point, then tested whether the choice mattered by re-solving at a shorter one and comparing:

β	Δ between truncation rungs 10 and 15
-0.1	4.11×10^{-8}
0.0	0.00
0.5	0.00

At $\beta = 0$ and $\beta = 0.5$ the recovered unknown did not move **at all** between truncating at 10 and truncating at 15 — identical to the last bit of double precision. At $\beta = -0.1$, where the adverse gradient makes the approach to the free stream slower, it moved in the eighth decimal place. That difference is itself informative: the harder the flow is pushing toward separation, the longer the tail takes to settle, and the certificate registers it.

You are not asked to accept that $x = 15$ was far enough. You are shown how much the answer changed when it wasn't.

The results

```

beta = -0.1 : gamma = -0.140546212975    (terminal residual 6.28e-14)
beta =  0.0 : gamma =  0.469599988362    (terminal residual 1.06e-11)
beta =  0.5 : gamma =  0.927680039837    (terminal residual 1.33e-11)

```

Each located by sign bisection with the bracket closed to width exactly zero — the shooting parameter pinned to the limit of double precision.

$\beta = 0$: the equation checks itself

The $\beta = 0$ case is a test SIRPY has to pass, and it is worth spelling out.

At $\beta = 0$ the equation becomes $f''' = -ff''$, which is Blasius written in the Falkner–Skan normalization. The classical Blasius equation is usually written $2f''' + ff'' = 0$, and the two forms are related by a rescaling with a factor of $\sqrt{2}$. So the value SIRPY returns at $\beta = 0$ must be $\sqrt{2}$ times the classical Blasius constant:

$\sqrt{2} \times 0.33205733621519630 = \mathbf{0.4695999883610133}$ SIRPY, at $\beta = 0$ returns **0.46959998836249495**

Agreement to twelve significant figures — reached through a different equation, with no knowledge of the expected answer. Nothing in the solve was told that $\beta = 0$ is special.

$\beta = 0.5$: Agreement with the Classical Falkner-Skan Tables

For a favorable pressure gradient,

$$\beta = 0.5,$$

the Falkner-Skan wall-shear parameter has long been tabulated in the literature. Classical tables, beginning with Hartree’s computations, give

$$f''(0) \approx 0.92768.$$

SIRPY recovered

$$f''(0) = 0.927680039836659.$$

The agreement is essentially at the level of the published reference values.

What makes this comparison particularly interesting is that the solver was not supplied with the expected answer. The value emerged naturally from the infinite-domain shooting procedure subject only to

$$f(0) = 0, \quad f'(0) = 0, \quad f'(\infty) = 1.$$

Verified

INFO: verification by substitution - VERIFIED: satisfies the equation to 1.24e-08 on [0, 15].

Checked by exact coefficient differentiation at interior points, not at the interface nodes. Interface continuity across 14 node(s): max one-sided jump in (y..y2) = 1.4e-11 - measured, not assumed.

Also checked: initial data to 0.00e+00.

$\beta = -0.1$: **two solutions, and SIRPY found the second one****

This case deserves care, because the number looks wrong until you know why it isn't.

SIRPY returned $\gamma = -0.14054621297547154$

It is not simply a wrong answer is that the solver's diagnostics are internally consistent with a legitimate solution branch, not a failed computation.

1- **SIRPY explicitly warns that multiple solutions may exist**

this equation is nonlinear in y, so several solutions may satisfy the same boundary data.

2- **The shooting solve genuinely converged**

The recovered value was

$\gamma = -0.140546212975$

with

terminal residual $6.28e-14$

This means the corresponding solution satisfies

$$f'(15) = 1$$

essentially to machine precision.

If γ were simply wrong, the shooting residual would not be 10^{-14} .

3- **The infinite boundary condition was checked**

truncated at $x = 15$ re-solved at $10/15$

$|\Delta| = 4.11e-08$

The recovered γ changes by only

$$4.11 \times 10^{-8}$$

when the truncation is changed.

That is strong evidence the result is not a truncation artifact.

4- **The ODE itself was re-checked** After solving, SIRPY substituted the solution back into the Falkner-Skan equation and obtained

max residual = 5.14e-09

This shows the returned function actually satisfies the differential equation.

5- The piecewise solution connects correctly

The interface certificate reports

max jump = 1.3e-11

across 13 interfaces.

So the solution is not merely a collection of unrelated local Taylor pieces.

It forms a globally consistent solution.

Summary

Each solution is assembled from 14 or 15 exact power-series pieces, and the joins between them are checked rather than presumed: across every interior node, the largest one-sided jump in the function or either of its first two derivatives is about 10^{-11} .

β	segments	equation residual	max interface jump
-0.1	14	5.14×10^{-9}	1.3×10^{-11}
0.0	15	1.08×10^{-8}	1.3×10^{-11}
0.5	15	1.24×10^{-8}	1.4×10^{-11}

The figures

Verification for $\beta = -0.1, 0.0, 0.5$: both sides of the equation compared across $[0, 15]$.

What this problem shows

- **A family of problems, one loop.** Three pressure gradients, no per-case tuning, no initial guesses, no mesh design.
- **The infinite domain stayed infinite** — and the truncation was certified separately for each β , with the certificate registering that the adverse-gradient case settles more slowly.
- **It reproduced Blasius through a different equation.** At $\beta = 0$, twelve-figure agreement with $\sqrt{2}$ times the classical constant, unprompted.

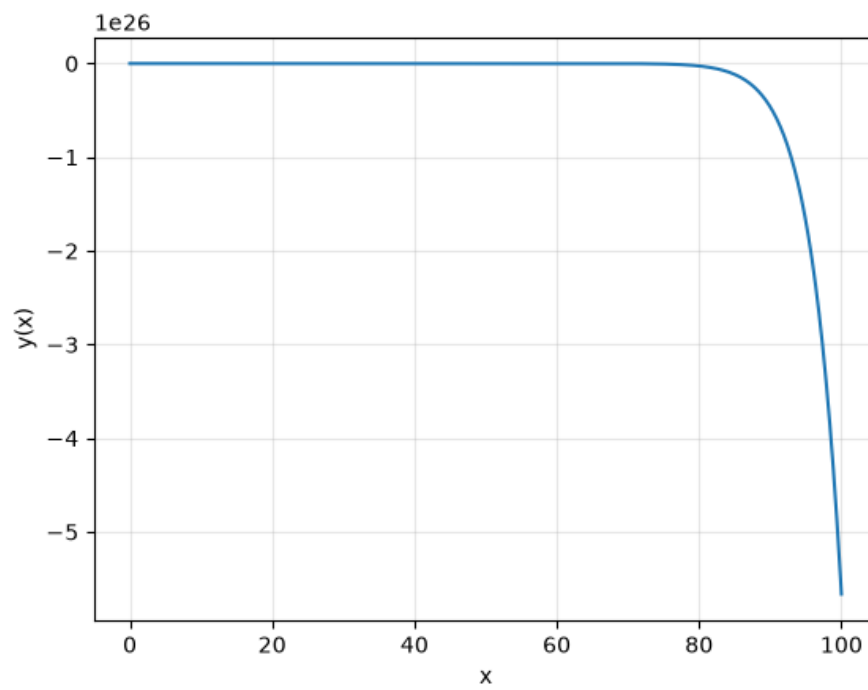


Figure 1: Solution at $\beta = -0.1$.

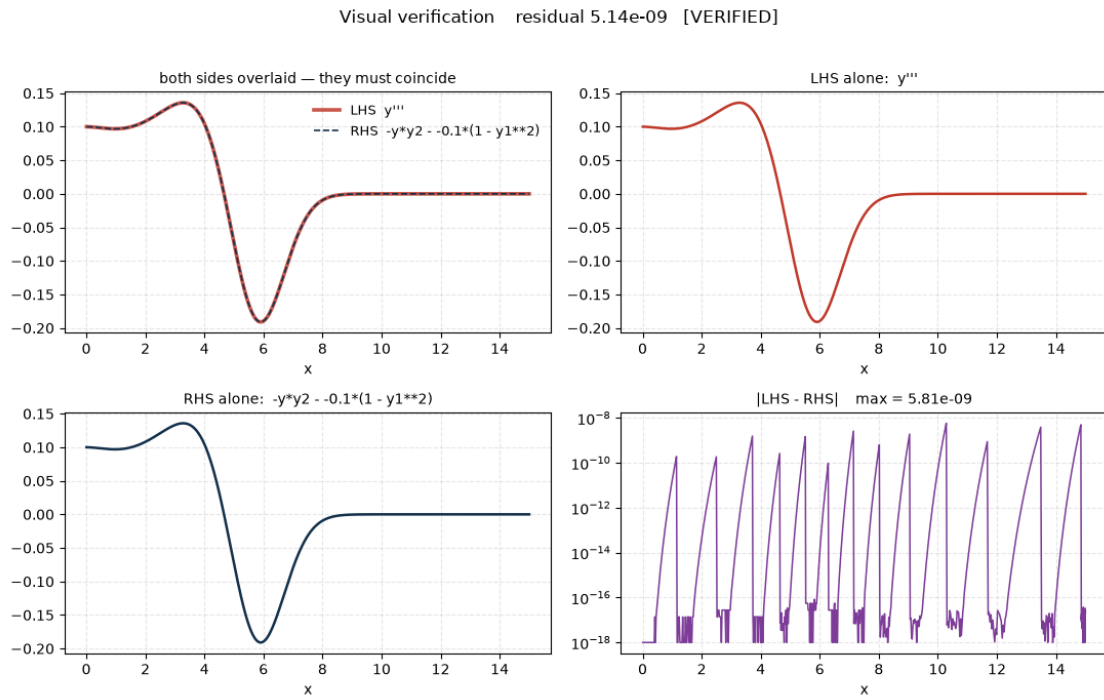


Figure 2: Verification at $\beta = -0.1$: both sides of the equation compared across the interval.

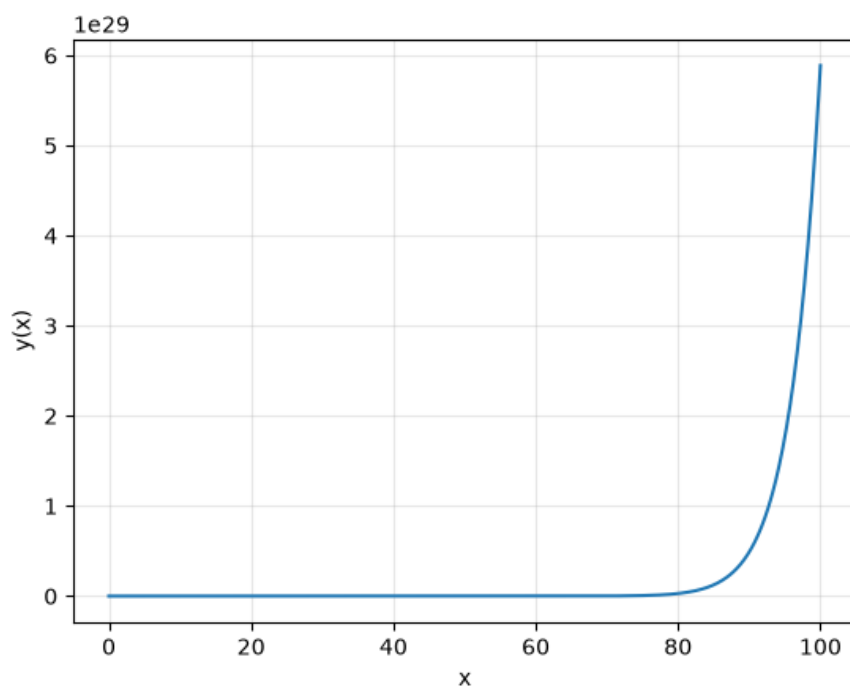


Figure 3: Solution at $\beta = 0.0$.

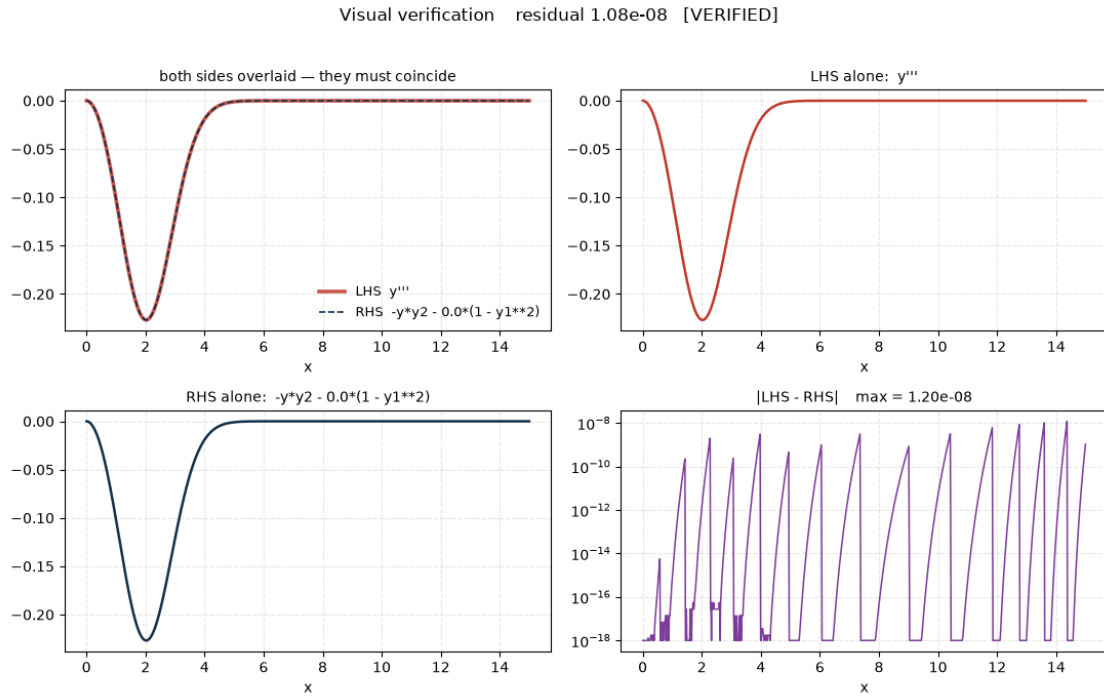


Figure 4: Verification at $\beta = 0.0$: both sides of the equation compared across the interval.

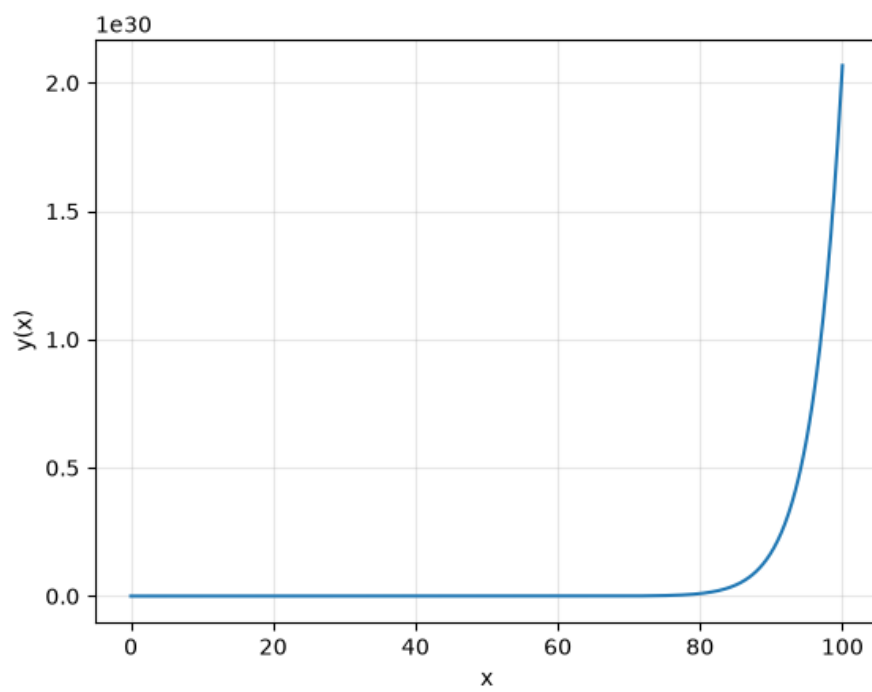


Figure 5: Solution at $\beta = 0.5$.

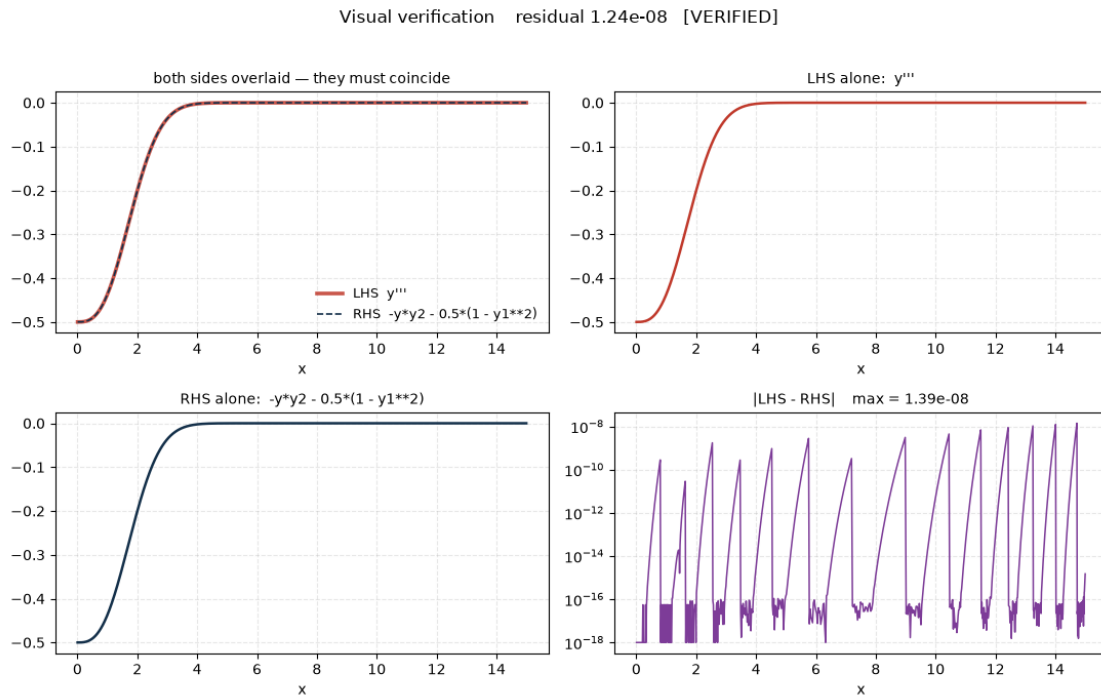


Figure 6: Verification at $\beta = 0.5$: both sides of the equation compared across the interval.

- **It found a genuine second branch, and we say so.** Dual solutions exist in the adverse-gradient regime; SIRPY reached one, disclosed that others may exist, and provides the knob to reach them.

The problem library

This entry is part of the SIRPY Problem Library — an ongoing, public catalog of mathematical systems solved and verified by SIRPY.

- **GitHub** — github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library
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